

convolutional neural networks
defects fft haadf-stem
stem denoising
afm mcp feature extraction
microscopy
4d-stem contrastive learning resnet electron microscopy pca
machine learning
segmentation explainable ai
deep learning super-resolution materials science
physics-guided machine learning image analysis ilm neural network
computer vision automated microscopy atomic-scale defects
sem digital twin defect classification hrtem
foam stability 2d materials preprocessing